

Lesson Notes

MODULE 4 | LESSON 1

RISK MODELING WITH EXTREME VALUE THEORY

|                 |                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reading Time    | 20 minutes                                                                                                                                                             |
| Prior Knowledge | GARCH, VaR, Expected Shortfall, PDF, CDF                                                                                                                               |
| Keywords        | Peaks Over Threshold (POT), Block Maxima Method (BMM), Extreme Value Theory (EVT), Generalized Pareto Distribution (GPD), Generalized Extreme Value (GEV) distribution |

In the last module, we applied different Deep Learning technologies to the challenge of Value at Risk estimation. In this module, we learn how to use Extreme Value Theory to better fit our empirical data to well-understood distributions in order to enhance our VaR estimates.

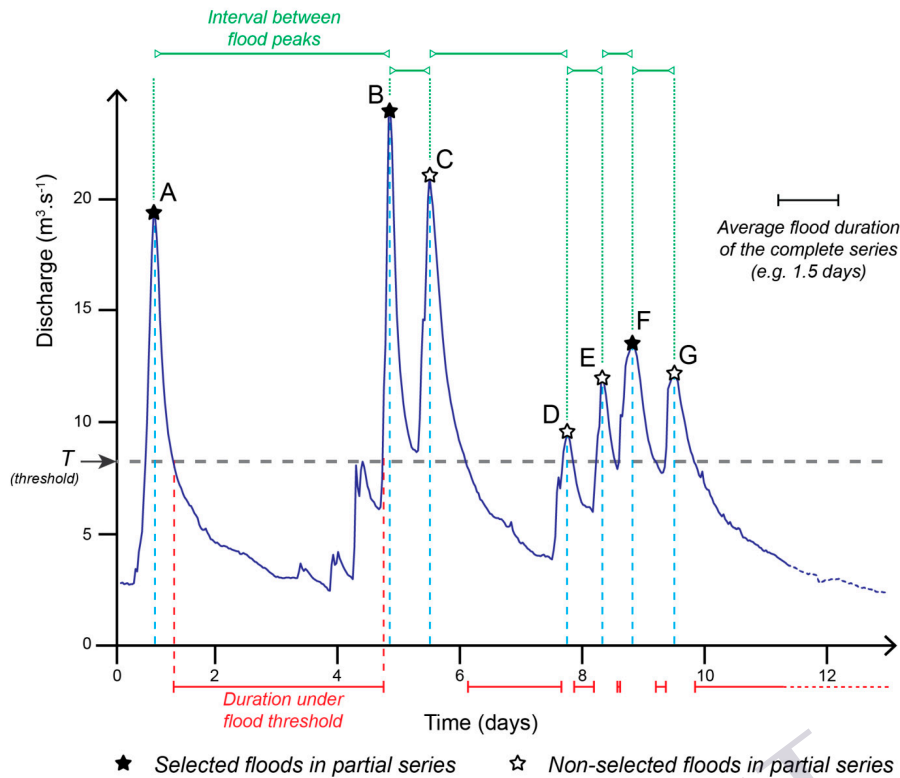
In Lesson 1 of Module 3, we discussed at length the significance of exceeding the Value-at-Risk (VaR) threshold. We reminded ourselves that such exceedences are actually built in to an informed use of VaR. As discussed in those lesson notes, we would not call such threshold exceedences “violations”; still, due to their magnitude and relative infrequency, we can aptly describe such PnL observations as “extreme.” We also recall that financial time series often exhibit what we call fat tails, meaning that the normal distribution that we find so tractable in most situations fails to account for the magnitude and frequency of these extreme values. Lucky for us, there is an entire theory devoted to modeling these extreme values, which we can make use of to better model (and so better prepare for) those PnL observations that exceed the VaR threshold. In this module, we learn about this theory and explore its techniques so we know better what to expect on the other side of our VaR threshold.

1. Peaks Over Threshold (POT)

In the language of the Peaks Over Threshold (POT) approach, the set of the exceedences are known as excesses, and these excesses over a given threshold in fact follow (in the limit) their own distribution, that is, the Generalized Pareto Distribution (GPD). In Section 2.1 of the required reading for this lesson, you find a precise mathematical formulation of the GPD, and you see that it takes two parameters,  $\xi$  for shape and  $\sigma$  for scale (Singh et al. 1480–1481). (An interesting fact: If the shape and scale are both zero, the GPD is equivalent to the exponential distribution.) Finding the values of these two parameters such that the resulting GPD fits the shape of our empirical data is clearly crucial. The authors mention using the typical approach to this parameterization: They find the maximum likelihood estimates, or MLE (Singh et al. 1481); we see how easy it is to perform these calculations in Python in the next lesson, where we also practice finding the appropriate parameters using the Hill’s estimator and the moment estimator.

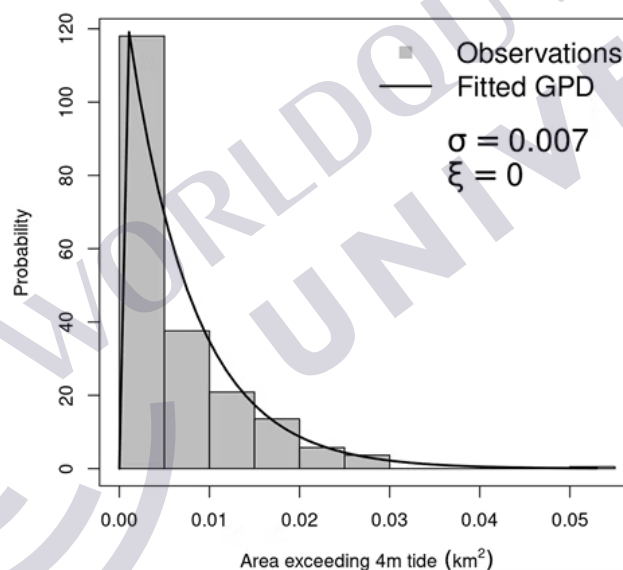
Figure 1 below illustrates, for a non-financial time series example, the selection of data to be modeled in POT: simply those observations in excess of the given threshold. Figure 2 below it sorts the peaks data (for a separate example) into a histogram, which shows how well the fitted GPD approximates it.

Figure 1: Peaks over Threshold



Source: van Campenhout, Jean et al. "Return Period of Characteristic Discharges From the Comparison Between Partial Duration and Annual Series, Application to the Walloon Rivers (Belgium)." *Water*, vol. 12, no. 3, 2020, <https://www.mdpi.com/2073-4441/12/3/792>

**Figure 2: Example of Peaks Fitted to GPD**



Source: Perera, Gonzalo, and Angel M. Segura. "Peaks over Manifold (POM): A Novel Technique to Analyze Extreme Events over Surfaces." *Advances in Pure Mathematics*, vol. 12, no. 1, 2022, <https://www.scirp.org/journal/paperinformation.aspx?paperid=114919>

Given the precise CDF for the GPD, we can straightforwardly estimate a VaR for a given quantile or "confidence level," and Singh et al. describe the relevant transformations and derivations simply and concisely. This implies that the threshold we use to define our excesses as described above does not need to match our VaR threshold, though we initially made the simplifying assumption above that VaR exceedances were the same as excesses. In fact, we might still choose to set our POT threshold to the .95 or .99 quantile, but using POT, we can evaluate VaR at even higher quantiles, for example, .999. It may also be the case that we find an appropriate threshold for our POT analysis is the .90 quantile (as do the authors of the paper), and by using POT (that is, taking advantage of the better fit between our empirical data and the properly parameterized GPD), we arrive at more accurate 95% and 99% VaR estimates.

Another advantage of a precise CDF for a good-fitting GPD is the ease with which we can calculate Expected Shortfall (ES), in many ways a much more informative risk measure. As you recall, VaR tells us (with more or less accuracy) that we should expect to exceed some loss some small percentage of the time, e.g., 5% or 1% of the time. However, without any information about the size of the losses to expect on these days, we really do not have enough information to adequately prepare for them. ES, on the other hand, fills in this missing information, and ES based on POT should provide more reliable estimates.

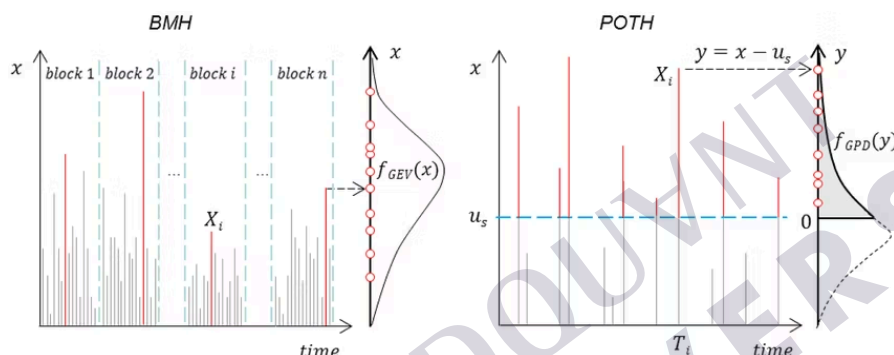
## 1.1 Dynamic EVT VaR and ES

The main contribution of Singh et al. is the innovative combination of modeling with POT *and* GARCH. Where previously, the POT approach to VaR and ES would use a static standard deviation, the authors propose regularly updating the volatility estimate assuming a GARCH(1,1) process. Given our knowledge that volatility is itself stochastic (the property known as “heteroscedasticity” which provides the last letter in GARCH), such evolving volatility measures have long been used in VaR and ES; however, extending these GARCH-VaR and GARCH-ES models to incorporate the additional precision of the GPD is somewhat newer—and an improvement both theoretically and empirically.

## 2. Block Maxima Method

EVT also includes the Block Maxima method (BMM), which “consists of dividing the observation period into non-overlapping periods of equal size” and takes the maximum observation in each period as its focus (Ferreira 1). POT considers all extreme observations, where extreme corresponds to the values from high or low quantiles of a relatively exhaustive distribution; on the other hand, BMM selects the (single) maximum from each of several “blocks” and fits them to the Generalized Extreme Value (GEV) distribution. Figure 3 illustrates this difference with a side-by-side comparison. Note the distribution specification in the graph along the right hand side of each of these. Do not be confused by the “H” in “BMH” and “POH” in the figure: It only indicates that these approaches rely on historical data.

**Figure 3: Visual Comparison of POT and BMM**



Source: Hamdi, Yasser et al. "Use of Historical Information in Extreme-Surge Frequency Estimation: The Case of Marine Flooding on the La Rochelle Site in France." *Natural Hazards and Earth System Sciences*, vol 15, iss. 7, <https://nhess.copernicus.org/articles/15/1515/2015/nhess-15-1515-2015.html>.

In fact, the GEV is actually a family of distributions, consisting of the Gumbel, Frechet, and Weibull distributions (Singh et al. 1479). See the [second required reading](#) for this lesson for an overview of each of the distributions in the GEV family, complete with insightful examples, helpful derivations, and thought-provoking comparisons.

It is generally accepted that POT is more efficient than BMM as POT makes “better use of the available information (Ferreira 2), but there are cases in which BMM may nevertheless be a better choice for modeling extreme values. Perhaps the clearest reason to prefer BMM to POT is when “there may be short range dependence that plays a role within blocks but not between blocks”; periodicity and other violations of the standard requirement or assumption that time series be independently and identically distributed (iid) make BMM a natural choice (Ferreira 2).

As with POT, we will experiment with implementation (and parameterization) of BMM using Python in the next lesson.

## 3. Conclusion

This was our introduction to the use of EVT in risk modeling. In the next lesson, we experiment in Python with the concepts touched on here. With the deeper understanding that comes with that hands-on modeling, we will then continue to learn how EVT is applied to risk measurement in various ways and markets.

### References

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